

Hein Zarni Aung

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Education

Stony Brook University

Expected Graduation: May 2027

Major: Computer Science + Applied Math & Statistics (Data Science)

GPA: 3.40

Honors and Scholarship Awards: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Software Security, Programming Abstractions, Systems Fundamentals, Algorithms, Statistics & Probability

Research

Ethos Lab @ Department of Computer Science

Stony Brook University

Undergraduate Research Assistant

Sep 2025–Present

- *Developing fraud detection algorithms to identify EZ-Pass scam patterns, analyzing payment processing vulnerabilities and transaction anomalies*
- *Automated scammer behavior monitoring using Selenium and Python, generating test payment data to estimate potential financial damage and identify attack vectors.*

Personal Projects

TuneAcademy – (Python, FastAPI, Docker, React, TypeScript, GCP, Essentia, Spotify Basic Pitch, NumPy)

- Built an AI-powered music education platform that records student audio, runs it through Spotify Basic Pitch for note transcription, and uses Essentia to extract pitch accuracy, rhythm consistency, and tonal quality — generating a structured weakness profile to match students with the right instructor.
- Containerized the Python/FastAPI backend with Docker, deployed on GCP with Cloud Firestore for real-time data sync; won Best Prenora Dynamics Hack at Hack Brooklyn 2026.

Discord Voice Channel Activity Monitor – (Node.js, Discord.js, Cloud)

- Built a Node.js/Discord.js bot with JSON-based settings and PM2 deployment on a cloud server to track user activity in voice channels.
- Implemented customizable DM notifications, VIP lists, and channel thresholds for each user and server with real-time monitoring.

Professional Experience

Department of Computer Science

Stony Brook University

Teaching Assistant – Data Structures & Algorithms

Aug 2025–Present

- Guided 30+ students in mastering asymptotic analysis, recursion, and key data structures (linked lists, trees, heaps, hash tables, graphs) through office hours and recitations.

- Conducted comprehensive midterm review sessions, creating practice problems and visual explanations to clarify challenging concepts.

Game Development and Design Club

Stony Brook University

Club Treasurer

Aug 2025 - Present

- Oversaw all budgeting and financial planning for club events, workshops, and project showcases, ensuring full compliance with university fiscal policy.
- Helped develop a game in Godot to be used in hands-on workshops to teach members foundational game design and development techniques. ([Github Link](#))

Office of Global Affairs

Stony Brook University

International Peer Mentor

Aug 2025 –Present

- Mentored a cohort of 7 international students through biweekly one-on-one meetings, providing academic guidance and cultural adjustment support tailored to individual needs.

Skills

Languages: Python, JavaScript, **TypeScript**, Java, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

ML & Data: NumPy, Pandas, Essentia, Spotify Basic Pitch

Databases & Cloud: MongoDB, PostgreSQL, Firebase, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium